INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SRI CITY

MID EXAMINATION –Spring 2026

Basic Electronics Circuits

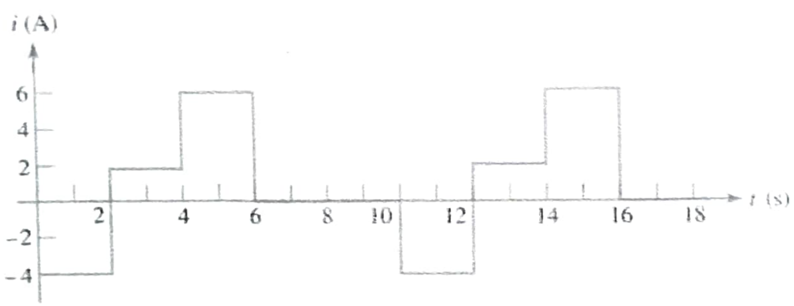
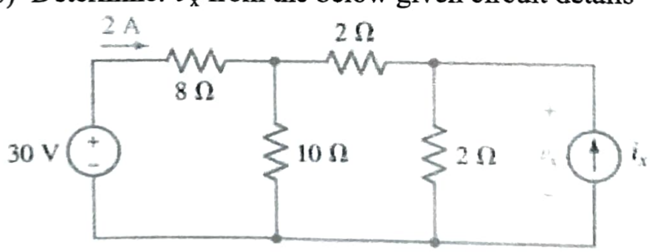
UG1 (PC) **Date: 24-02-2026**

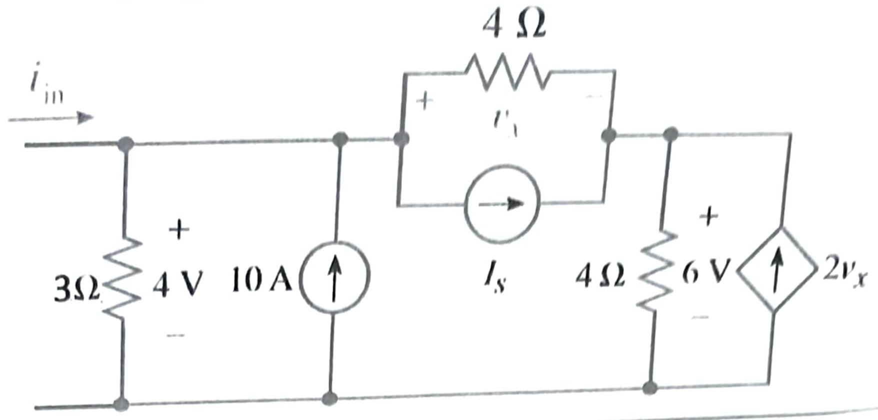
Duration: 90 Mins(2:30 to 4 PM) **Max. Marks: 25**

Instructions: **Roll I**

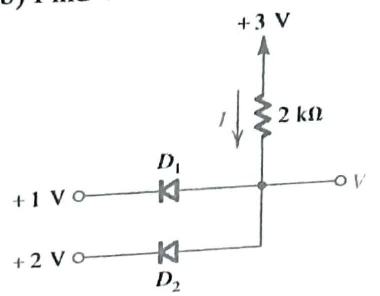
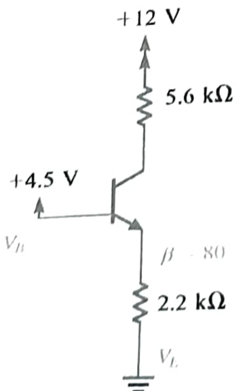
1. Closed book exam
2. Calculators are allowed. Sharing in exam hall is strictly not allowed
3. Assumptions made should be clearly stated
4. All sub-parts of the question should be written together
5. Question paper must be attached with the answer booklet

Section-A

1	a) Figure Shown below has a period of 10 seconds. i)What is the average value of the current over one period ii) How much charge is transferred in the interval 1<t<12 ?  	2 M
	b) Determine: v_x from the below given circuit details  	2 M
	c) Define Voltage	1M
2	a)Derive L_{eq} for a parallel combination of Inductors	**1.5 M**
	b) Use Ohm's law and Kirchoff's laws on the circuit below and find i) v_x ii) i_{in} iii) Power provided by the dependent source and justify	**3.5 M**



Section-B

<table>
<tr><td>3</td><td>a) Explain PN junction with a neat diagram indicating Majority, Minority and bound charges, Depletion region indicating the Diffusion and Drift currents for forward bias of the Diode</td><td>2 M</td></tr>
<tr><td></td><td>b) Find V and I from the below circuit assuming Diodes are ideal

</td><td>1 M</td></tr>
<tr><td></td><td>c) Explain BJT in Common Emitter configuration and Active mode with a Diagram Explaining the minority and majority charge distributions at Emitter-Base Junction and Collector-Base Junction</td><td>2 M</td></tr>
<tr><td>4</td><td>a) Comment on the mode of the circuit given below to determine the voltages at all nodes and the currents through all branches, Assume that β is specified to be 80
V_{CC}=+12 V, V_B=+4.5 V, R_C=5.6 kΩ, R_E=2.2 kΩ , V_{BE}=0.7 V

</td><td>3M</td></tr>
</table>

	b) Explain briefly NPN amplifier and Voltage-Transfer Characteristics and derive $i_c = g_m v_{be}$ where $g_m = \frac{I_C}{V_T}$	2M
5	a) Derive the expression of Ripple voltage for a shunt capacitor filter	2M
	b) Using a Diode bridge circuit with RC filter and a Zener diode will be in parallel with the output load. An ac input voltage with RMS value 230 V and at 60 Hz is available. Zener diode with Vzo = 15 V (Iz = 50 mA) and resistance of 3 ohms. And a transformer with an 10:1 turns ratio is available. Diode cut-in voltage = 0.8 V, Ripple voltage = 1.5 V, RL = 600 ohms, and R = 250 ohms. Determine, Vs (peak), Vo1 (voltage across the capacitor), Vo.	3 M

************************** All the Best **********************************